

DEPARTMENT OF COMPUTER SCIENCE & STATISTICAL LEARNING

Course: CS402 - Machine Learning & Statistical Inference

Unit 1: Normal Distribution & Standard Deviation \u03c3 Analysis

1.1 Core Mathematical Definitions

The standard deviation \u03c3 measures the amount of variation or dispersion of a set of values.

A low standard deviation \u03c3 indicates that the values tend to be close to the mean \u03bc of the set.

The variance \u03c3² is defined as the expectation of the squared deviation of a random variable X .

In Gaussian probability density functions, \u03c3 determines the width of the bell curve.

Formally, for a random variable X with mean \u03bc, the variance is $\u03c3^2 = E[(X - \u03bc)^2]$.

The cumulative distribution function satisfies $\Phi(x) \rightarrow 1$ for all real x .

1.2 Feature Scaling and Standardization

Standardization transforms features to have zero mean \u03bc = 0 and unit variance \u03c3 = 1.

The z-score transformation formula is $z = (x - \u03bc) / \u03c3$.

When \u03c3 > 0, scaling ensures that features contribute equally to distance metrics in k-NN and SVMs.

If \u03c3 = 0, the feature exhibits zero variance and must be pruned prior to model training.